Franklin Osualaaham

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EDUCATION

University of Texas at Arlington — Arlington, TX
Bachelor of Science in Data Science | Expected May 2027
Coursework: Machine Learning, Statistical Inference, Data Mining, Applied Statistics, Data Visualization

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, NumPy), SQL, R, SAS, statistical analysis
Excel & Visualization: Excel (PivotTables, VLOOKUP, formulas), Tableau, Power BI
Databases & Processing: PostgreSQL, MySQL, SQLite, PySpark
Machine Learning & AI: scikit-learn, XGBoost, PyTorch, Keras, graph features, OCR, RAG, LlamaIndex
Development: Git/GitHub, Streamlit, Gradio, Docker fundamentals

RELEVANT EXPERIENCE

University of Texas at Arlington — Softball | Student Data Science Contributor | Aug 2026 - Present

- Analyzed pitch-level tracking data using Python to compare pitcher and batter profiles and explore player-evaluation scenarios.
- Developed Tableau dashboards and presented findings to coaches and classmates, contributing to discussions about extending the analysis across a full season.
- Identified missing-data limitations and recommended tracking and collection improvements to support reliable player comparisons within program budget constraints.

NY Racing Team | Data Science Extern, Partnerships Team | Jun 2026 - Aug 2026

- Integrated race results, television exposure, and social-media engagement using Python and web scraping, reducing estimated Excel-based reporting preparation from three days to one.
- Built an integrated MySQL database; used CTEs, window functions, joins, and subqueries to prepare sponsorship data for analysis.
- Developed a weighted sponsorship scoring model using exposure and engagement proxies to compare four partners; visualized results in Tableau to support renewal discussions.
- Presented findings to partnership leadership, with recommendations shared with ownership to inform sponsorship-renewal strategy.

Pfizer | Applied AI Extern, Clinical Supply Chain Team | Jan 2026 - Apr 2026

- Built an AI document-processing prototype using Python and OCR to extract and organize searchable information from pharmaceutical vendor documents with inconsistent layouts.
- Extended the prototype with a LlamaIndex-based RAG workflow and Gradio interface, enabling natural-language questions across uploaded documents.
- Evaluated OCR and retrieval approaches across varied document formats; documented findings and recommended automation, scaling, and deployment improvements with senior data scientists and AI engineers.

SELECTED PROJECTS

FraudGraph — Bitcoin Transaction Risk Analysis | GitHub

- Investigated whether network relationships improve illicit-transaction identification, engineering graph features across 203,769 Bitcoin transactions and comparing XGBoost models using time-based splits.
- Improved average precision by 23.6% relative to the transaction-only baseline; a top-5% review queue captured 42.9% of labeled illicit test cases. Later-period performance declined.

Retail Marketing Intelligence — Promotion Evaluation | GitHub | Dashboard

- Analyzed a randomized email experiment with 64,000 customers using Python and SQL; estimated purchase-rate increases of 0.68 and 0.31 percentage points versus no email.
- Built a Streamlit dashboard comparing campaign effects and targeting strategies; held-out evaluation found no clear advantage for uplift-based targeting over random selection.

Anfield Matchday Intelligence — Ticket Utilization | GitHub | Website

- Combined ticketing reports and fixture records for 38 Liverpool FC home league matches across two seasons using Python and SQL to investigate unused tickets and supporter access.
- Built a React dashboard exploring ticket patterns and hypothetical recovery scenarios; on nine held-out fixtures, historical-average forecasts outperformed Ridge regression (MAE: 437 vs. 454 tickets).

Facial Expression Recognition — Senior Capstone | Ongoing

- Developing a four-class classifier for angry, happy, sad, and neutral expressions, comparing traditional machine-learning models with CNN-based approaches.
- Evaluating models for a planned live-camera and image-upload application, with AI-agent integration enabling questions about live classification results.